

DATA MINING

Introduction
to Statistics

SENTIMENT
ANALYSIS

Numerical
Analysis

Text Mining

NLP

Introduction to Text Mining

analysis opinion
data text data
processing language
analysis

Dr. Farman Ali

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Department of Applied AI

SUNGKYUNKWAN UNIVERSITY

Lecture-1

Mining Models Retrieval Semantic Web Extraction Classification Link Mining Text Analysis Information Learning

natural language processing NLP learning interaction

Farman Ali



- Associate Professor at the Department of Applied AI, Sungkyunkwan University, South Korea
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- +82-10-9224-0977
- EB/AE: 4 Journals
- Top 2% Researcher, 2021~latest
- Listed Best Scientist on research.com

Publications / Citations / Impact Factor

Total Articles Published in Journals 190+

Articles Published in JCR top 10% Journals 80+

Under Review Articles in Journals and Conf 25+

Registered Patents 04

Conference Papers 13

Book Chapters 02

Total Citations 10100+

H-index 56

Total Impact Factor 900+

Participation in Research Projects 8

Professional and Research Experience 16+

Research Interests

❖ Data Mining and Social Networking Analysis

- Text mining and topic modeling.
- Information extraction from textual data.
- Social networking analysis for healthcare systems.
- Social networking analysis for ITS.

❖ Medical Informatics

- Healthcare monitoring system for chronic patients.
- AI approaches for disease detection.
- Machine learning for medical diagnosis.
- Analysis of electronic medical records.

❖ Medical LLM Systems

- Multilingual natural language processing.
- Mental health text analysis.
- Explainable AI for healthcare applications.
- Clinical decision support systems.

- **Staff**

- Farman Ali (farman0977@g.skku.edu or farmankanju@gmail.com)

- **Lecture Location & Time**

- Tuesday 12:00 ~13:15 and Thursday 13:30~14:45
 - Room #: 9B215 , International Hall

- **Grading Policy**

Midterm(25%), Final exam(40%), Attendance (10%), Presentation+ Assignments(25%)

- **Cheating Policy**

- Automatic **F** for both



Course Description: This course provides an introduction to the fundamental concepts, techniques, and applications of text mining and sentiment analysis. It offers a clear overview of how to extract valuable insights and patterns from unstructured textual data. Students will explore key topics, including natural language processing (NLP), text preprocessing, various levels of natural language, text analysis techniques, text classification and evaluation, document classification, sentiment analysis, topic modeling, word embeddings, and language models. The course emphasizes practical learning through hands-on exercises and real-world case studies, enabling students to grasp technical concepts and their applications across different fields.

Course Notes:

- Lectures will be conducted for 3 hours per week.
- Office hours: (Room: 90312, International Hall)
 - Tuesday 14:00 ~ 17:00
 - Thursday 15:00 ~ 17:30
- Class activities and Quizzes will be used for continuous assessments.
- Notes will be uploaded to (<https://icampus.skku.edu/>)

Textbook:

An Introduction to Text Mining: Research Design, Data Collection, and Analysis (Gabe Ignatow, University of North Texas, USA)

Additional Guide:

- Lecture materials and assignments will be based on content drawn from recommended books and the internet.
- Quizzes will be based on the material delivered during the lecture.
- The lecture material will be presented in the form of PowerPoint slides or handout notes.
- Both PPT slides and a whiteboard will be used during the lecture to discuss topics and solve equations/problems. Slides and handout notes will be provided before the lecture time.

The classes will be face-to-face



Course Syllabus

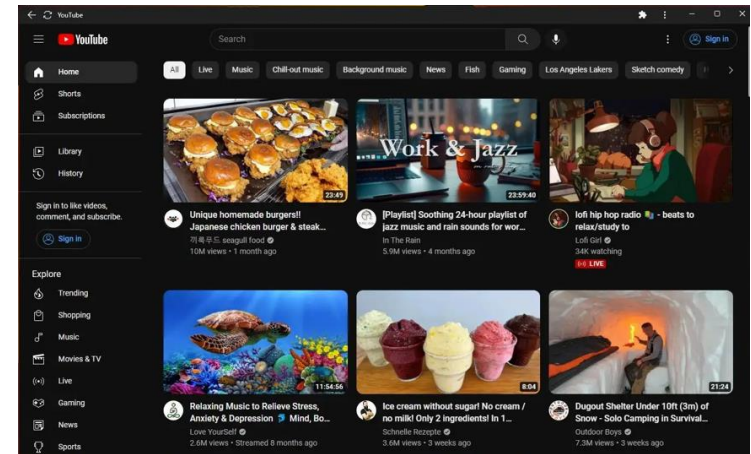
- Introduction to the course
- Levels of Natural Language
- Installation of R Libraries and Text Preprocessing
 - Sentiment Analysis Using R
 - Text Representation Using R
 - Challenges in Lexicon-based Text Classification
 - Naïve Bayes and SVM-based Text Classification Using R
 - The Rise of Deep Learning Approaches
 - Topic Modeling Using R
 - Presentation of Research papers about Text Mining

What is Data? What is Structured and Unstructured data?

Data can be represented in various types of structures, formats, and media.

CUSTOMER

CUSTOMER_ID	LAST_NAME	FIRST_NAME	STREET	CITY	ZIP_CODE	COUNTRY
10302	Boucher	Leo	54, rue Royale	Nantes	44000	France
11244	Smith	Laurent	8489 Strong St	Las Vegas	83030	USA
11405	Han	James	636 St Kilda Road	Sydney	3004	Australia
11993	Mueller	Tomas	Berliner Weg 15	Tamm	71732	Germany
12111	Carter	Nataly	5 Tomahawk	Los Angeles	90006	USA
14121	Cortez	Nola	Av. Grande, 86	Madrid	28034	Spain
14400	Brown	Frank	165 S 7th St	Chester	33134	USA
14578	Wilson	Sarah	Seestreet #6101	Emory	1734	USA
14622	Jones	John	71 San Diego Ave	Arlington	69004	USA



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statistics

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Statistics Abadir, Karim M/ Cambridge, UK Samsung Library [519.5 A116s] Available Western Book

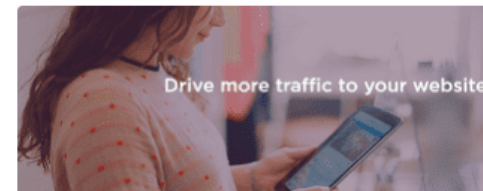
Statistics :concepts and controversies Moore, David S/ New York, NY :W. H. Freeman,c2017. Samsung Library [519.5 M821s9] Available Western Book

Statistics :an introduction using R Crawley, Michael J/ Chichester, West Sussex :John Wiley & Sons, Inc.,c2015. Samsung Library [519.502855133 C911s2] Available Western Book



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Drive more site traffic with a website card campaign.

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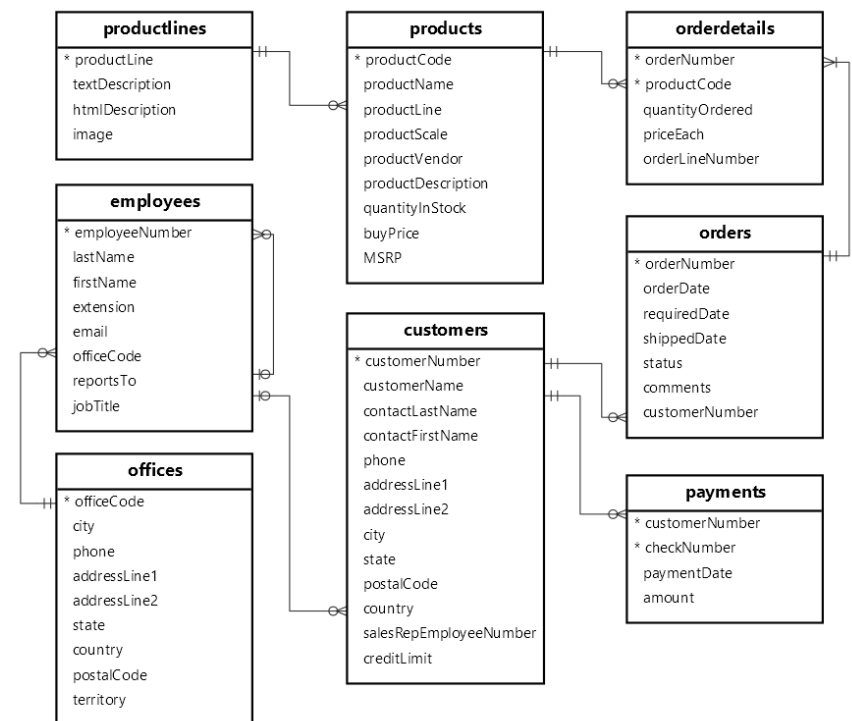
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Structured Data refers to data with a high level of organization, such as in relational databases and spreadsheets.

- Depends on a data model- a model of the data
- Types and how they will be stored, processed and accessed
- Easily entered, stored, queried and analyzed.
- Structured Query Language is used for management of structured data.



Example of MySQL sample Database

Unstructured Data means all things that cannot be classified and fit into one simple model.

- Photos and graphic images
- Videos
- Streaming instrument data
- Webpages, emails, blog entries, wikis
- PDF files, and ppt and doc documents



Structured and Unstructured Features in Social Media Data



Social media data contains both **structured metadata** and **unstructured text**.
Text mining helps us extract meaningful knowledge from this combination.



STRUCTURED FEATURES

Easy to measure and quantify

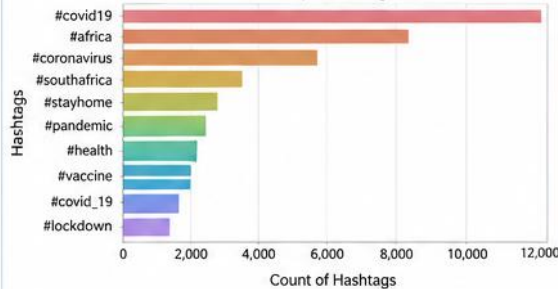
	What?	Type of post, day, time
	Where?	Location, place
	Who?	User ID, age, gender
	How many?	Likes, shares, retweets, comments
	How often?	Hashtag count, posting frequency
	When?	Timestamp, duration

Best for **quantitative analysis**

Numbers, counts, trends, comparisons

EXAMPLE VISUALIZATION

Top Hashtags Overall



Helps answer
**WHAT, WHEN,
WHERE, HOW MANY,
and HOW OFTEN.**

VS.



UNSTRUCTURED FEATURES

Rich in meaning, context, and opinion

	What is being said?	Post content, comments, replies
	How does the user feel?	Sentiment, emotion, opinion
	What is it about?	Topics, themes, subjects
	Who/what is mentioned?	Entities, people, organizations
	What is the context?	Context, intent, motivation
	Why is it important?	Insights, narratives, perspective

Best for **semantic / qualitative analysis**

Meaning, themes, sentiment, insights

EXAMPLE TEXT ANALYSIS



Alysha De Valle @ABC7Alysha · 3h

Traffic alerts cleared from lanes on 10 W, here's traffic on 10 EAST at Milliken Ave. Use 60 EAST instead. #IE #Traffic



eddie irvin @irvinedda · 1h

RT @TrafficLeicester: Accident at the junction of Regent Road, Accident Granville Road is closed due to the Accident outside.



Sarah Khan @sarahk · 2h

Heavy traffic on the highway this morning because of road work. Plan extra time!

Sentiment	Topic	Entities	Intent
Positive	Traffic Update	10 W, 10 EAST, Milliken Ave.	Inform
Negative	Accident	Regent Road, Granville Road	Inform / Alert
Negative	Traffic, Road Work	Highway	Inform

Extracts meaning, opinion, topics, entities and intent



KEY TAKEAWAY:

Structured features tell us the facts.
Unstructured text tells us the **meaning** behind the facts.

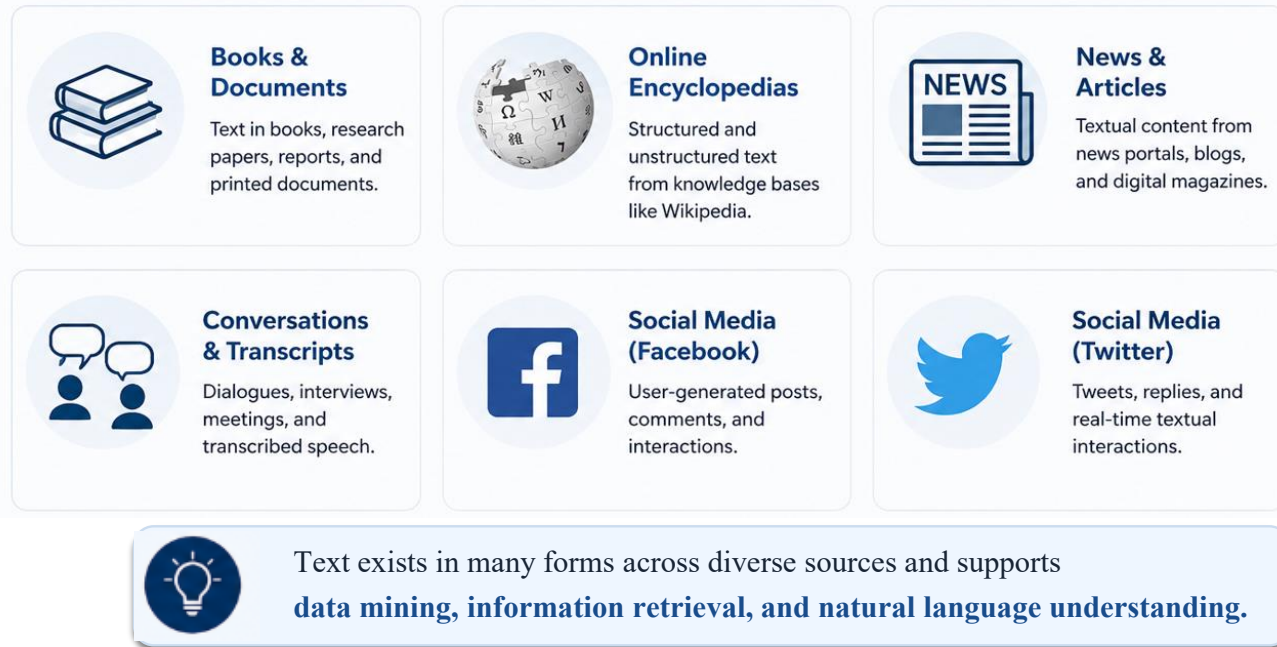


Combining both leads to
deeper understanding and **better decisions**.

What is Text?

Text refers to any **written or transcribed language** produced by humans, which can be processed and analyzed to extract information and knowledge.

Where does text come from?



Data is growing fast!

> **7 PB** of data generated every second worldwide $\approx$ 600+ PB per day

Source: IDC Global DataSphere Forecast (2025–2029)

The challenge is not just collecting data — **it is extracting useful knowledge from it.**

Volume of Unstructured Text Data?



Total Volume (2025):

150–300 ZB

(zettabytes) of unstructured text exists globally.



Daily Growth:

10–30 PB

(petabytes) per day
(~10–30 million TB daily)

Unstructured Text Data by Source (Estimated Daily Volume)

Source	Daily Volume (PB)	Examples
Emails	~3–6 PB/day	Business, personal
Social Media (Text)	~2–5 PB/day	Tweets, Reddit, Facebook posts
Business Documents	~2–4 PB/day	PDFs, Word, Excel, PowerPoint
Chats & Messaging	~1–3 PB/day	WhatsApp, Slack, SMS
Web Content	~1–2 PB/day	Blogs, forums, news
Logs & Telemetry	~1–5 PB/day	Server logs, app analytics

Example: A Tweet / Social Media Post



United Nations @UN

Palestinians are enduring a humanitarian catastrophe + of horrific proportions. Children in Gaza are dying from starvation and bombardments. We need unimpeded humanitarian access across Gaza. Now. #Gaza #UNRWA

10:15 AM · May 10, 2024 · X for iPhone

364

1.2K

3.4K

215K



Author

United Nations (@UN)



Description

Official statement on the humanitarian situation in Gaza.



Location

New York, USA
(From profile metadata)



Social Network

X (formerly Twitter)
Public Post

What can be mined from the text?



TOPIC

What is it about?



SENTIMENT

Positive, negative, neutral, or emotion



AUTHOR

Who produced it?



TIME

When was it created?



LOCATION

Where is it relevant?



POPULARITY

How much engagement?

What is Text Mining?



Text Mining, or Text Data Mining, or approximately Text Analytics, is the process of deriving **high-quality information or actionable knowledge** from texts.

Text Mining helps us:



Minimizes Human Effort

Automates extraction and analysis of large volumes of text, reducing manual work and time.



Supplies Knowledge for Optimal Decision Making

Transforms raw text into meaningful insights that support better, data-driven decisions.

From Text to Insight



1. Text Collection
Gather text from multiple sources.



2. Preprocessing
Clean, normalize, and prepare the text.



3. Text Mining
Extract patterns, topics, and relationships.



4. Knowledge Discovery
Derive meaningful information and insights.



5. Decision Making
Apply insights to make smarter decisions.

Core Components of Text Mining



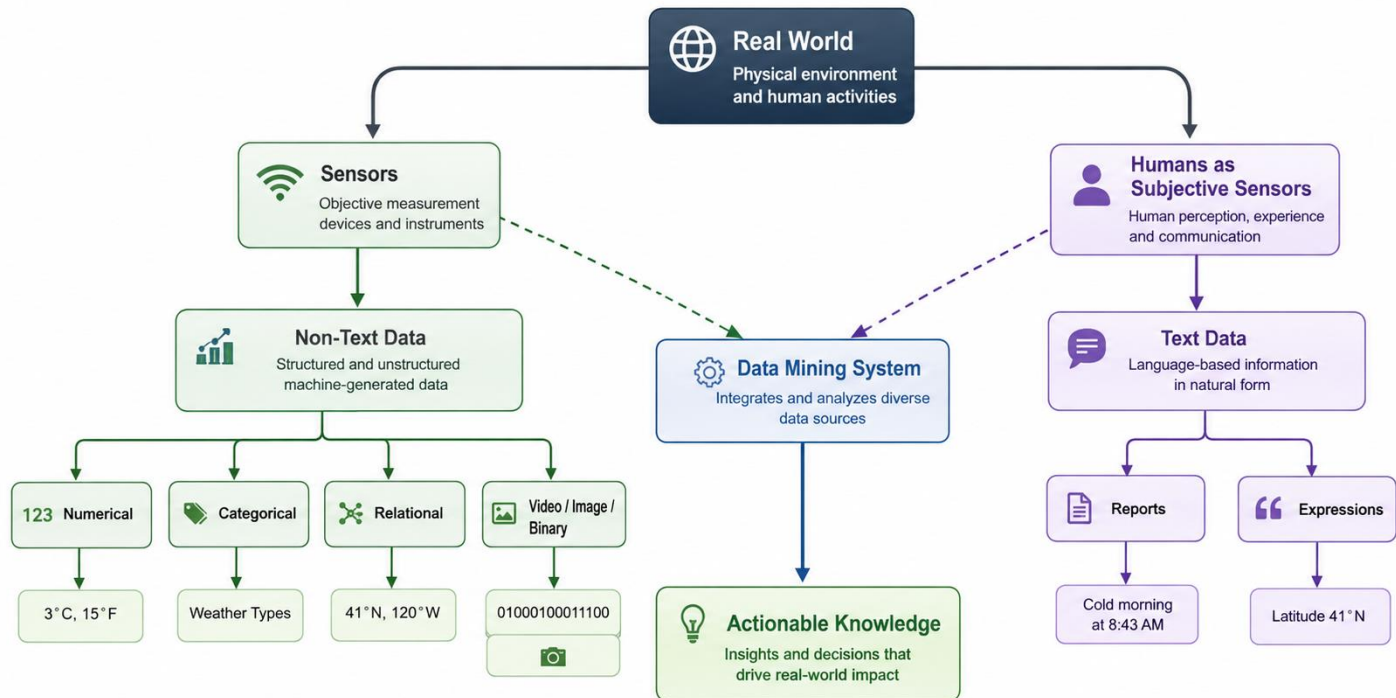
Data Mining

- We will be dealing with both text and non-text data.
- Extracts patterns from structured data (numbers, categories, tables).

Scenario: A supermarket analyzes sales data.

Input: "Customer A bought milk + cereal on Monday"

Output: Discovers *"60% of customers who buy milk also buy cereal"*



Text Mining

- An extension of Data Mining (which analyzes numerical data) to textual data.
 - Extracts meaning from unstructured text (natural language).
- Goal:** Discover hidden patterns, trends, or insights from large text collections (e.g., social media posts, research papers).
- Input:** *Input:* 10,000 product reviews
- Output:** Identifies trends (e.g., "80% of reviews mention 'battery' negatively").

Web Mining

- Extracts patterns from web data (both structured and unstructured).
- In Text Mining, the **input** is free **unstructured text**, while web sources are structured.

Information Retrieval (Information Access)

- Finds existing information without creating new knowledge
- Input:** Your query.
- Output:** Returns links to articles/blog posts that *already exist*.

Computational Linguistics (CL)

- Uses statistical methods on large text datasets to solve language-related problems.

Goal: Provide data-driven foundations for NLP tasks.

Example: Computes how often the word "bank" refers to a riverbank vs. a financial institution in a corpus.
Informs algorithms for Word Sense Disambiguation (determining a word's meaning in context).

Natural Language Processing (NLP)

- Applies CL insights to build practical tools that understand/generate human language.

Goal: Solve real-world problems like translation, sentiment analysis, or chatbots.

Example: Uses CL's POS tagging (identifying nouns/verbs) to improve grammar-checking software.

Statistics

- This provides the **mathematical framework for validation**. It offers the tools to measure the reliability, significance, and confidence of the patterns discovered through text mining.

Text Mining

It deals with the text itself

- Frequency count of words
- Sentence length
- Word spectrum
- Presence/absence of key words
- etc.

CL and NLP

It deals with the underlying metadata

- Linguistics
- Computational linguistics
- Content analysis
- Stylistics
- etc.

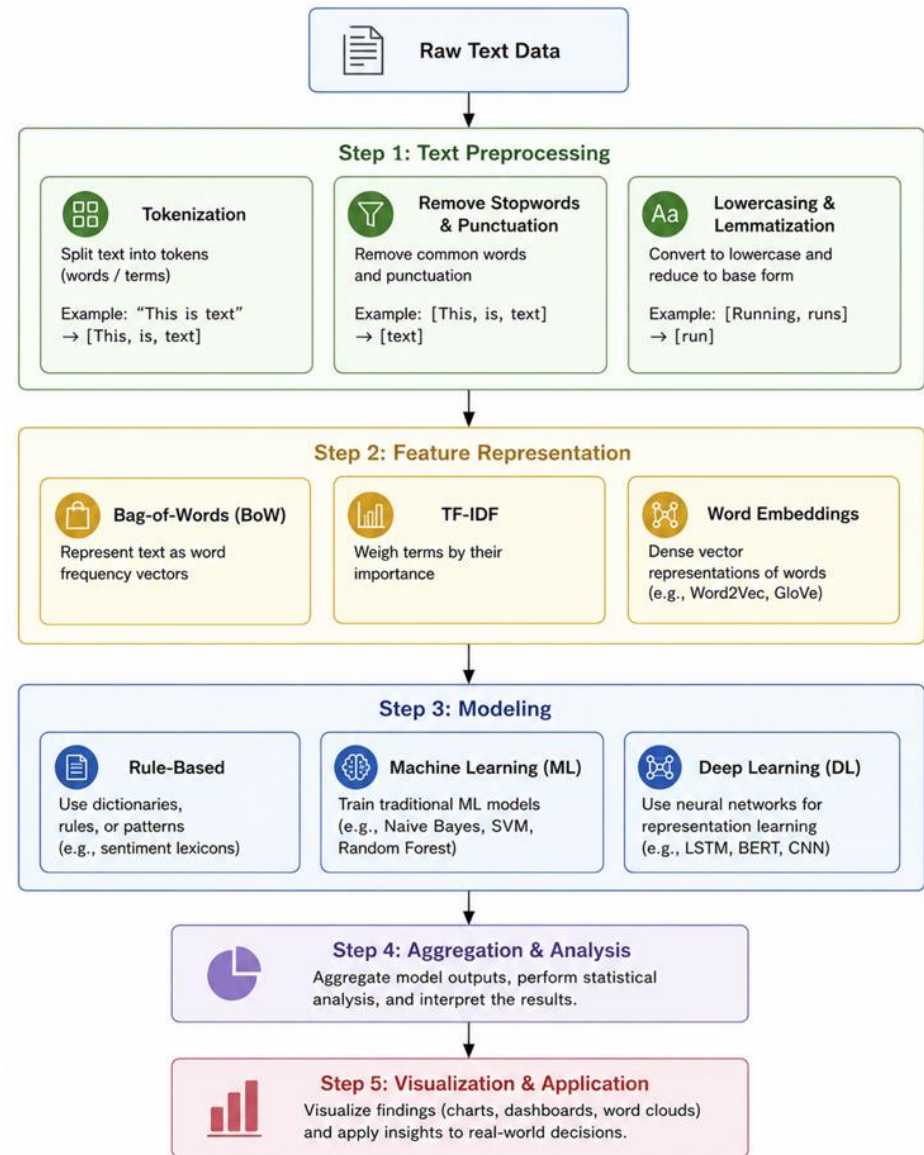
Roughly summed up:

CL analyzes text to discover language rules

NLP uses these rules to build tools

Text Mining applies NLP/CL techniques to extract business insights.

- Sentiment Analysis
- Information Extraction
- Text Classification
- Text Summarization
- Text Clustering
- Topic Modeling



Sentiment Analysis

The goal of sentiment analysis is to determine the polarity of a given text at the document, sentence, or feature/aspect level

Polarity values

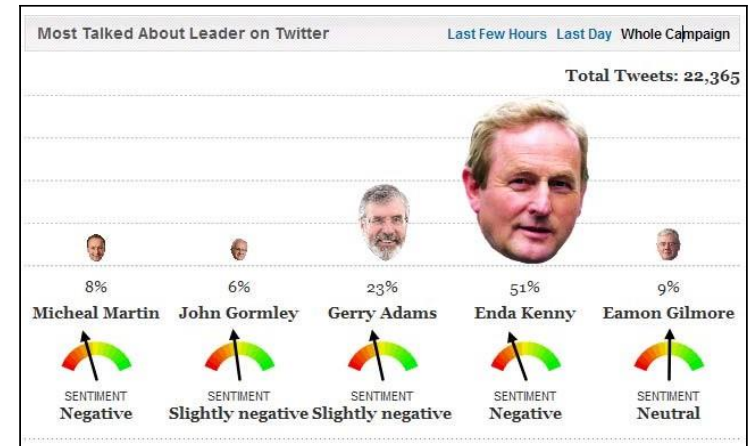
- Positive, neutral, negative or Likert scale (1 to 10)

Application examples

- Document level
 - Analysis of tweets about politicians
- Feature/aspect level
 - Analysis of product reviews

Approaches

- Rule-Based → Lexicons
- Machine Learning → learn patterns



Corpus E

d_1 (Document 1)

The cat sat on the mat. The cat was happy.

1 2 3 4 5 6 7 8 9 10

d_2 (Document 2)

The dog played in the park.

1 2 3 4 5 6

Final $TF \times IDF$

The final measure of a word's importance to a document.

$$TF-IDF("the", d_1) = 0.30 \times 0 = 0$$

$$TF-IDF("the", d_2) = 0.33 \times 0 = 0$$

$$TF-IDF("cat", d_1) = 0.20 \times 0.69 = **0.138**$$

Calculate the TF-IDF for the word **"the"** in both documents.

Term Frequency (TF) Calculation

How important a word is within a single document.

Term	In Document	Count	Total Terms	TF (Count / Total)
"the"	d_1	3	10	0.30
"the"	d_2	2	6	~0.33
"cat"	d_1	2	10	0.20

Inverse Document Frequency (IDF) Calculation

How unique or rare a word is across the entire corpus.

Term	Docs With Term	Total Docs	IDF ($\log(\text{Total Docs} / \text{Docs With Term})$)
"the"	2	2	$\log(2/2) = 0$
"cat"	1	2	$\log(2/1) = \sim 0.69$
"dog"	1	2	$\log(2/1) = \sim 0.69$

The IDF score of 0 perfectly captures that this word is not a good discriminator between documents and holds no special "relevance" or "importance" for any specific document in the corpus.

Information extraction

- Identification of entities, their attributes, and their relations in text.
- **Example.** Extract company's founding dates from news articles

Possible text analysis steps

1. Lexical and syntactic preprocessing
2. Named and numerical entity recognition
3. Reference resolution
4. Entity relation extraction

Time entity **Organization entity**
" 2014 ad revenues of Google are going to reach

Reference **Time entity**
\$20B. The search company was founded in '98.

Reference **Time entity** **Founded relation**
Its IPO followed in 2004. [...] "

Output: **Founded("Google", 1998)**

Information Extraction

- Information extraction is the task of automatically extracting structured information from unstructured or semi-structured documents.

Subtasks

- Named Entity Recognition and Disambiguation
 - “The parliament in Berlin has decided ...”
 - Which parliament? Which Berlin?
- Relationship Extraction
 - PERSON works for ORGANIZATION
 - PERSON located in LOCATION
- Fact Extraction
 - CITY has population NUMBER
 - COMPANY has turnover NUMBER [Unit]

small dogs - Google Squared

http://www.google.com/squared/search?q=small+dogs

Web Images Maps News Video Gmail more

Google squared small dogs Square it Add to this Square

Sign in to save your Squares

Item Name	Image	Description	Weight	Height	Group
Chihuahua		Do not let the Chihuahua get away with things you would not allow a large dog to do (Small Dog Syndrome), such as jumping up on humans. ...	6 lb	6-9 inches	Toy
Maltese		Do not allow these dogs to develop Small Dog Syndrome, human induced behaviors. ... Today, the glamorous Maltese is an adored pet and sought-after show dog. ...	8 lbs	18 to 24"	Toy
Bichon Frise		A Bichon Frise (French, literally meaning curly lap dog) is a small breed of dog of the Bichon type. They are popular pets, similar in appearance to, ...	7-12 lbs	9 - 12 inches	Non Sporting
Affenpinscher		Description, The Affenpinscher is a small dog with a harsh, shaggy coat, and longer hair all over the face. It is a smaller version of a working terrier and ...	7-10 lbs.	10 - 15 inches	Toys
Brussels Griffon		Brussels Griffon Breed Standard. Toy Group. General Appearance A toy dog, intelligent, alert, sturdy, with a thickest, short body, a smart carriage and ...	6-12 pounds	7-8 inches	Toy
Pomeranian		You may find Pomeranian puppies for sale and Pomeranian dogs for sale from quality dog ... She is So Small. She Has A Gorgeous Little Baby Doll Face. ...	4lbs	7-12 inches	Toy
Havanese		The Havanese gives a rugged impression of a little dog. It is sturdy, and while a small breed, it is neither fragile nor overdone. ...	7-13	8-11	Toy
Australian Terrier		"Dedicated to the Advancement of Quality, Purebred Australian Terriers". Founded	12 - 14 lb	10 in	Terrier

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Results 7 to 12 of 12

Your Filters Reset Filters

item type Skyscraper (12) floor count 50 and up (1) building completed in year up to 2000 (1) location Hong Kong (1)

Highcliff

Highcliff is a 252.4-metre (828-foot) tall skyscraper located on a south slope of Happy Valley on the Hong Kong Island in Hong Kong. The 75 storey (70 floors of which are livable space) building's construction began in 2000 and was completed in 2003 under a design by DLN Architects & Engineers. It was the Silver Winner of the 2003 Emporis Skyscraper Award, coming in second to 30 St Mary Ave in London.

The Harbourside

The Harbourside is a 255 m (836.6 ft) tall residential skyscraper located at 1 Austin Road West, in Union Square complex on Kowloon peninsula. The building is erected on the West Kowloon Reclamation west of Kwan Chung. Construction of the 74 storey building began in 2000 and was completed in 2003 under the design by P & T Architects & Engineers. The building is, in fact, three towers joined at the base, middle

Text Classification

- Assignment of a text to one class from a given set of classes.
- **Example.** Classify topic and sentiment of customer reviews

*“ This was truly a lovely hotel to stay in .
The staff were all friendly and very helpful .
The location was excellent . The atmosphere is
great and the decor is beautiful. ”*

Possible text analysis steps

1. Lexical and syntactic preprocessing
2. Feature computation and supervised classification

Notice

- Applications often combine information extraction and text classification.

Output: Topic("hotel"), Sentiment("positive")

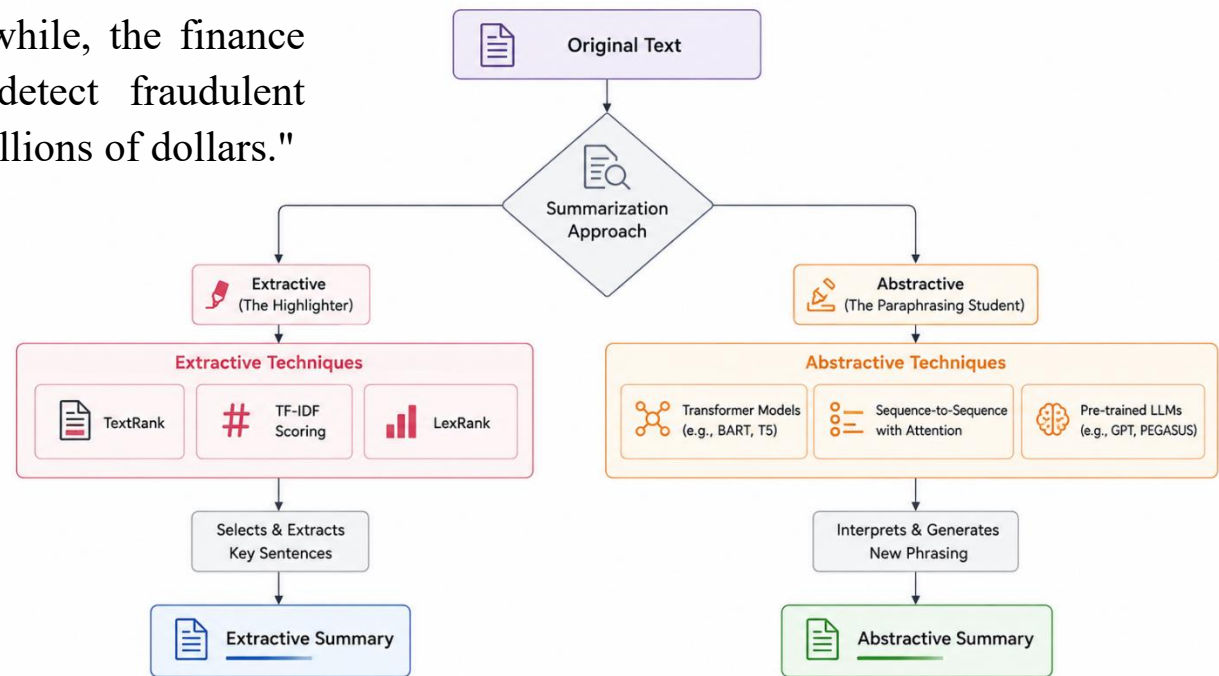
Text Summarization

- The process of automatically creating a shorter version of a text document that retains the most important information and overall meaning.

"Advances in artificial intelligence are transforming many industries. In healthcare, AI is being used to analyze medical images for early disease detection with remarkable accuracy. Meanwhile, the finance sector uses AI algorithms to detect fraudulent transactions in real-time, saving billions of dollars."

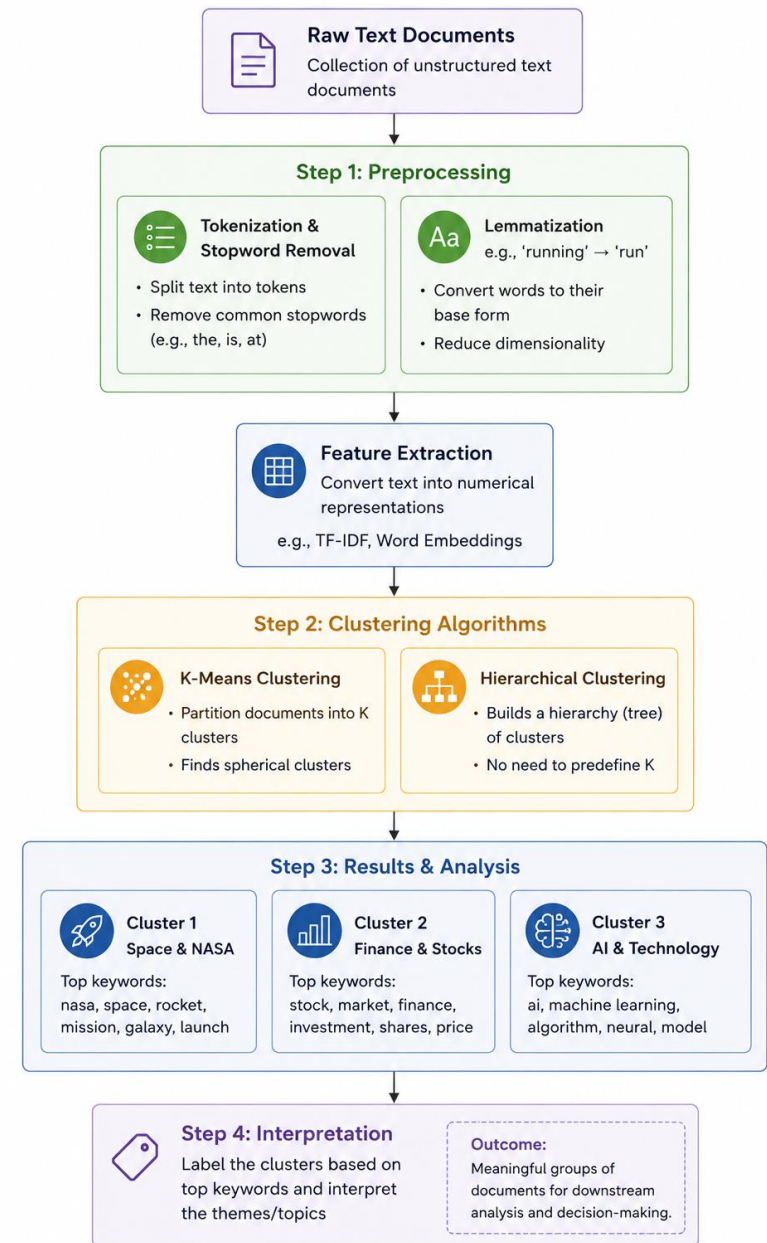


In healthcare, AI is being used to analyze medical images for early disease detection with remarkable accuracy. Meanwhile, the finance sector uses AI algorithms to detect fraudulent transactions in real-time, saving billions of dollars.



Text Clustering

- It is an unsupervised machine learning technique that automatically groups a collection of text documents (like articles, emails, or tweets) into meaningful categories, called clusters.



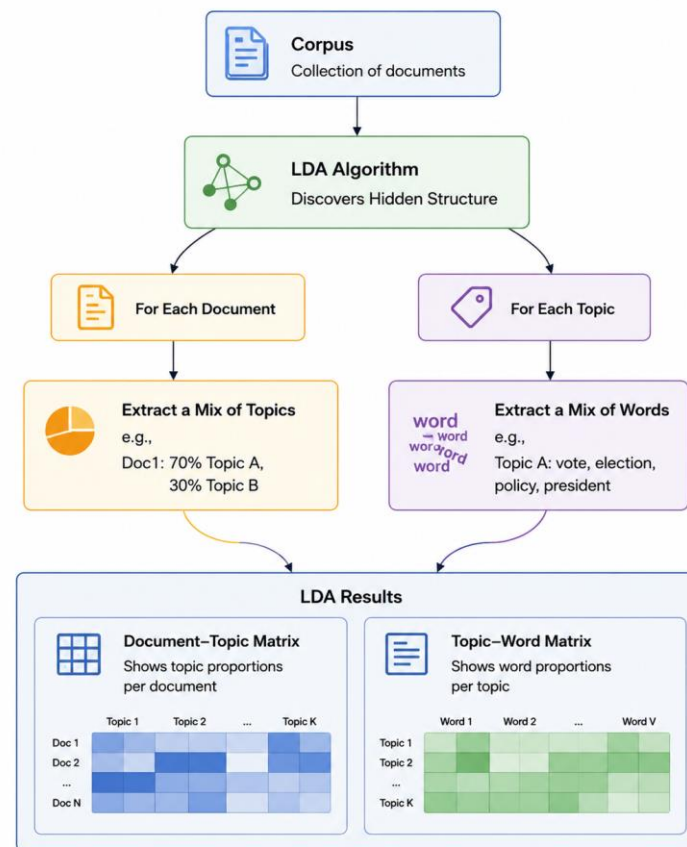
Topic Modeling

- Topic Modeling is an unsupervised machine learning technique that automatically discovers the hidden thematic structure in a large collection of documents. It's a form of text mining for finding and tracing patterns of words.

Latent Dirichlet Allocation (LDA)

	T1	T2		Tn
D1	1	0		0
D2	1	1		0

	W1	W2	...	Wn
T1	1	0		1
T2	0	1		0



- **Presentation:** Each student must present three or four papers published in top-tier conferences or journals related to any of the following topics.
- **Project:** Each student must present programming code for any of the following topics.

Topics

- Sentiment Analysis
- Information Extraction
- Text Classification
- Text Summarization
- Text Clustering
- Topic Modeling
- Word Embedding
- LLM

Programming: Visual Studio Code with extension Python or R



감사합니다